

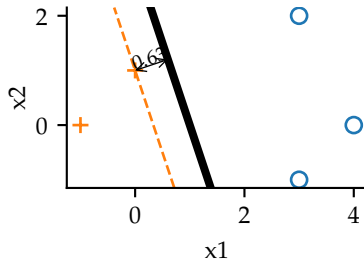
Chapter 7

Answer 7.1

To answer the first two parts, we can build the following table for $\vec{w} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ and $b = 3$.

x_1	x_2	y	$y_i(\vec{w}^\top \vec{x}_i + b)$	$\frac{y_i(\vec{w}^\top \vec{x}_i + b)}{\ \vec{w}\ }$
-1	0	+1	6	$\frac{6}{\sqrt{10}}$
0	1	+1	2	$\frac{2}{\sqrt{10}}$
3	2	-1	4	$\frac{4}{\sqrt{10}}$
4	0	-1	9	$\frac{9}{\sqrt{10}}$
3	-1	-1	7	$\frac{7}{\sqrt{10}}$

- a. From the table above, we can see that the minimum geometric margin is $\frac{2}{\sqrt{10}} \approx 0.63$. Although not necessary, we can plot this:

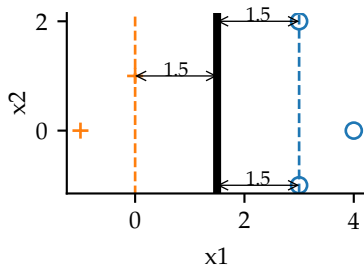


- b. Similarly, the minimum functional margin is 2.

To answer the next two parts, we can build the following table for $\vec{w} = \begin{bmatrix} -6 \\ 2 \end{bmatrix}$ and $b = 6$.

x_1	x_2	y	$y_i(\vec{w}^\top \vec{x}_i + b)$	$\frac{y_i(\vec{w}^\top \vec{x}_i + b)}{\ \vec{w}\ }$
-1	0	+1	12	$\frac{12}{\sqrt{40}}$
0	1	+1	4	$\frac{4}{\sqrt{40}}$
3	2	-1	8	$\frac{8}{\sqrt{40}}$
4	0	-1	18	$\frac{18}{\sqrt{40}}$
3	-1	-1	14	$\frac{14}{\sqrt{40}}$

- c. From the table above, we can see that the minimum geometric margin is $\frac{4}{\sqrt{40}} = \frac{2}{\sqrt{10}}$. This is the same as the previous answer because the line is in the same position.
- d. The minimum functional margin is 4 which is twice as large because both $\vec{w}$ and b are twice as large.
- e. A quick plot of the points shows that the maximum margin is 1.5:



Shifting or rotating this line will produce a smaller margin.

- f. Any scaling of $\vec{w} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ and $b = 1.5$ will produce this line ($x_1 = 1.5$). We know that functional margin needs to be 1 for the point $[0, 1]$. Therefore, letting β be the scaling,

$$\beta \left(\begin{bmatrix} -1 \\ 0 \end{bmatrix}^\top \begin{bmatrix} 0 \\ 1 \end{bmatrix} + b \right) = 1$$

$$\beta (0 + 1.5) = 1$$

$$\beta = \frac{1}{1.5}$$

and thus the parameters are

$$\vec{w} = \begin{bmatrix} -\frac{1}{1.5} \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} \\ 0 \end{bmatrix}$$

$$b = \frac{1}{1.5} = \frac{2}{3}.$$

Answer 7.2

To answer the first two parts, we can build the following table for $\vec{w} = \begin{bmatrix} -2 \\ -1 \\ 3 \end{bmatrix}$ and $b = -5$.

x_1	x_2	x_3	y	$y_i(\vec{w}^\top \vec{x}_i + b)$	$\frac{y_i(\vec{w}^\top \vec{x}_i + b)}{\ \vec{w}\ }$
-1	-1	3	+1	7	$\frac{7}{\sqrt{14}}$
1	-1	3	+1	3	$\frac{3}{\sqrt{14}}$
-1	0	4	+1	9	$\frac{9}{\sqrt{14}}$
0	0	2	+1	1	$\frac{1}{\sqrt{14}}$
1	2	0	-1	9	$\frac{9}{\sqrt{14}}$
1	1	0	-1	8	$\frac{8}{\sqrt{14}}$

- a. From the table above, we can see that the geometric margin is $\frac{1}{\sqrt{14}} \approx 0.27$.
b. Similarly, the functional margin is 1.

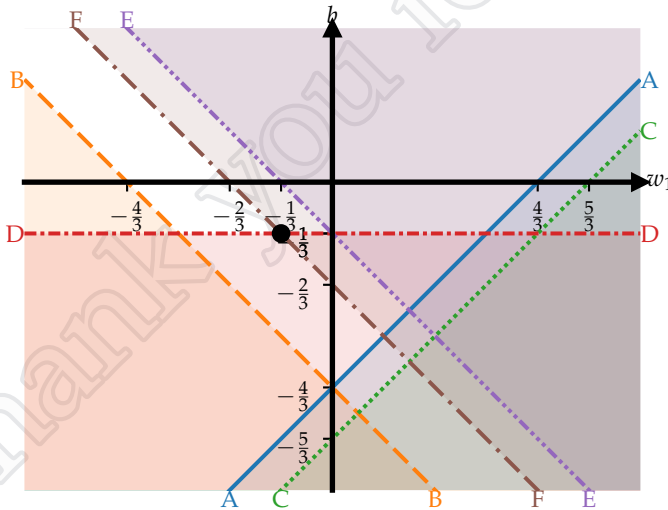
c. The data points induce the following constraints

$$\begin{aligned}
 A: & +1 \left(\begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}^T \begin{bmatrix} w_1 \\ -1/3 \\ 2/3 \end{bmatrix} + b \right) \geq 1 \\
 B: & +1 \left(\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}^T \begin{bmatrix} w_1 \\ -1/3 \\ 2/3 \end{bmatrix} + b \right) \geq 1 \\
 C: & +1 \left(\begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix}^T \begin{bmatrix} w_1 \\ -1/3 \\ 2/3 \end{bmatrix} + b \right) \geq 1 \\
 D: & +1 \left(\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}^T \begin{bmatrix} w_1 \\ -1/3 \\ 2/3 \end{bmatrix} + b \right) \geq 1 \\
 E: & -1 \left(\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}^T \begin{bmatrix} w_1 \\ -1/3 \\ 2/3 \end{bmatrix} + b \right) \geq 1 \\
 F: & -1 \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}^T \begin{bmatrix} w_1 \\ -1/3 \\ 2/3 \end{bmatrix} + b \right) \geq 1
 \end{aligned}$$

which simplify to

$$\begin{aligned}
 A: & -w_1 + b \geq -4/3 \\
 B: & w_1 + b \geq -4/3 \\
 C: & -w_1 + b \geq -5/3 \\
 D: & b \geq -1/3 \\
 E: & -w_1 - b \geq 1/3 \\
 F: & -w_1 - b \geq 2/3
 \end{aligned}$$

Constraint A dominates constraint C and constraint F dominates constraint E, so we do not need to consider C and E. However, we can plot them all:



The unshaded area is not forbidden by any constraint. SVM seeks the weight vector with the smallest magnitude. Because w_2 and w_3 are fixed, this amounts to picking the w_1 in the unshaded region

that is closest to $w_1 = 0$. That is the circled point, so we get the solution

$$\begin{aligned} w_1 &= -1/3 \\ b &= -1/3 \end{aligned}$$

Answer 7.3

In each case, the plot with the highest C value has the largest penalty on points that violate the margin. Therefore it will have the fewest points within the margin or misclassified. It also corresponds to the decision surface with the least regularization, so it will probably look the least “smooth.”

By contrast, the plot with the lowest C value will have the largest margin (although more points will be within the margin) and the “smoothest” decision boundary.

- (2) has the smallest C value and (3) has the largest C value.
- (1) has the smallest C value and (2) has the largest C value.
- (1) has the smallest C value and (3) has the largest C value.
- (2) has the smallest C value and (1) has the largest C value.

Answer 7.4

- The loss function is

$$L = \sum_{i=1}^m \max(0, 1 - y_i \vec{w}^\top \vec{x}_i) + \frac{1}{2C} \vec{w}^\top \vec{w}$$

which means the gradient is

$$\nabla_{\vec{w}} L = \sum_{i=1}^m -g(y_i \vec{w}^\top \vec{x}_i) y_i \vec{x}_i + \frac{1}{2C} \vec{w}$$

where

$$g(z) = \begin{cases} 0 & \text{if } z > 1 \\ 1 & \text{otherwise} \end{cases}.$$

Therefore, the update rule is

$$\vec{w} \leftarrow \vec{w} + \eta \left(\sum_{i=1}^m g(y_i \vec{w}^\top \vec{x}_i) y_i \vec{x}_i - \frac{1}{2C} \vec{w} \right).$$

- We can put the regularizer inside the sum by dividing it by m and adding it to each term in the sum over examples:

$$L = \sum_{i=1}^m \left(\max(0, 1 - y_i \vec{w}^\top \vec{x}_i) + \frac{1}{m2C} \vec{w}^\top \vec{w} \right).$$

Following the derivation above, the update rule for a single example is

$$\vec{w} \leftarrow \vec{w} + \eta g(y_i \vec{w}^\top \vec{x}_i) y_i \vec{x}_i - \frac{\eta}{2C} \vec{w}.$$

- c. This is very similar to the perceptron learning rule update with two exceptions. First, there is a weight decay term. Second, the example, $\vec{x}_i$, is added (for a positive example) or subtracted (for a negative example) if the discriminant times the label is less than 1 (instead of less than 0 for the perceptron). This means that examples that are correct, but not by “enough” still result in a correction.

Answer 7.5

- a. If a point violates the (hard) margin, the slack variable (ξ_i) must be positive to satisfy the constraint $y_i (\vec{x}_i^\top \vec{w} + b) \geq 1 - \xi_i$. If the point does not violate the margin, there is no benefit (to the constraints) for having a negative slack variable. Further, (unlike in the L1-SVM case) a negative slack variable will increase (not decrease) the objective, meaning that the program will naturally seek to keep the slack variable non-negative without any additional constraints.

The primal program is

$$\begin{aligned} \arg \min_{\vec{w}, b, \xi_1, \dots, \xi_m} \quad & \frac{1}{2} \vec{w}^\top \vec{w} + \frac{C}{2} \sum_{i=1}^m \xi_i^2 \\ \text{s.t.} \quad & y_i (\vec{x}_i^\top \vec{w} + b) \geq 1 - \xi_i \quad \text{for all } i \in \{1, \dots, m\} \end{aligned}$$

- b. The values for ξ_i can still be rewritten as $\xi_i = \max(0, 1 - y_i \tilde{f}(\vec{x}_i))$ which means the objective for an L2-SVM is

$$\frac{1}{2} \vec{w}^\top \vec{w} + \frac{C}{2} \sum_{i=1}^m \left(\max(0, 1 - y_i \tilde{f}(\vec{x}_i)) \right)^2$$

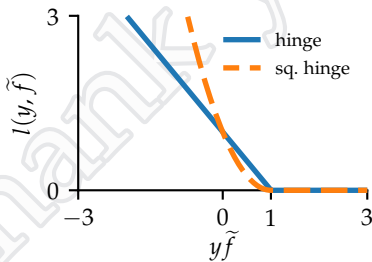
which is the same as minimizing

$$L = \sum_{i=1}^m l(y_i, \tilde{f}(\vec{x}_i)) + \lambda R(\vec{w})$$

where

$$\begin{aligned} l(y, \tilde{f}) &= \left(\max(0, 1 - y\tilde{f}) \right)^2 \\ R(\vec{w}) &= \vec{w}^\top \vec{w} \\ \lambda &= \frac{1}{C} . \end{aligned}$$

Plotted, the two hinge losses are



- c. The Lagrangian is

$$J = \frac{1}{2} \vec{w}^\top \vec{w} + \frac{C}{2} \sum_{i=1}^m \xi_i^2 - \sum_{i=1}^m a_i (y_i (\vec{x}_i^\top \vec{w} + b) - 1 + \xi_i)$$

which is the same as Equation 7.13 except that the last term has been dropped and the penalty on the slack variables (second term) has a square and a coefficient of $\frac{C}{2}$.

This yields the constraints at the solution of

$$\begin{aligned} 0 &= \nabla_{\vec{w}} J & 0 &= \frac{\partial J}{\partial b} & 0 &= \frac{\partial J}{\partial \xi_i} \\ &= \vec{w} - \sum_{i=1}^m a_i y_i \vec{x}_i & &= \sum_{i=1}^m a_i y_i & &= C \xi_i - a_i \\ \vec{w} &= \sum_{i=1}^m a_i y_i \vec{x}_i & 0 &= \sum_{i=1}^m a_i y_i & \xi_i &= \frac{a_i}{C} \end{aligned}$$

which are the same as in Equation 7.15, except for the right column.

Plugging the solution constraints back into the Lagrangian, we get

$$\begin{aligned} J &= \frac{1}{2} \vec{w}^\top \vec{w} + \sum_{i=1}^m \frac{C}{2} \xi_i^2 - \sum_{i=1}^m a_i y_i \vec{x}_i^\top \vec{w} - \sum_{i=1}^m a_i y_i b + \sum_{i=1}^m a_i - \sum_{i=1}^m a_i \xi_i \\ &= \frac{1}{2} \vec{w}^\top \vec{w} + \sum_{i=1}^m \frac{C}{2} \frac{a_i^2}{C^2} - \vec{w}^\top \vec{w} + \sum_{i=1}^m a_i - \sum_{i=1}^m \frac{a_i^2}{C} \\ &= -\frac{1}{2} \vec{a}^\top \mathbf{G} \vec{a} + \vec{a}^\top \vec{1} - \frac{1}{2C} \vec{a}^\top \vec{a} \end{aligned}$$

for the same definition of $\mathbf{G}$ as in the text.

Thus, as a minimization, the QP for L2-SVM is

$$\begin{aligned} \arg \min_{a_1, \dots, a_m} \quad & \frac{1}{2} \vec{a}^\top \mathbf{G} \vec{a} - \vec{a}^\top \vec{1} + \frac{1}{2C} \vec{a}^\top \vec{a} \\ \text{s.t.} \quad & 0 \leq a_i \quad \text{for all } i \in \{1, \dots, m\} \\ & \vec{a}^\top \mathbf{Y} = 0 \end{aligned}$$

Answer 7.6

a. The general form for a hypersphere is (for any a)

$$a(\|\vec{x} - \vec{c}\|^2 - r^2) = 0$$

which can be written as

$$a \vec{x}^\top \vec{x} + (-2a \vec{c})^\top \vec{x} + (a \vec{c}^\top \vec{c} - ar^2) = 0$$

which implies the features of $\vec{x}^\top \vec{x}$ (a scalar), $\vec{x}$ (a vector of features), and a scalar constant are required:

$$\phi(\vec{x}) = \begin{bmatrix} \vec{x}^\top \vec{x} \\ \vec{x} \\ 1 \end{bmatrix}$$

b. The inner product of two such constructed features vectors is

$$\begin{aligned}
 k(\vec{x}, \vec{x}') &= \begin{bmatrix} \vec{x}^\top \vec{x} \\ \vdots \\ \vec{x} \\ \vdots \\ 1 \end{bmatrix}^\top \begin{bmatrix} \vec{x}'^\top \vec{x}' \\ \vdots \\ \vec{x}' \\ \vdots \\ 1 \end{bmatrix} \\
 &= \vec{x}^\top \vec{x} \vec{x}'^\top \vec{x}' + \vec{x}^\top \vec{x}' + 1
 \end{aligned}$$

Answer 7.7

a. The inner product between $\phi(\vec{x})$ and $\phi(\vec{x}')$ is

$$\begin{aligned}
 k(\vec{x}, \vec{x}') &= 1 + x_1 x'_1 + x_2 x'_2 + \cdots + x_{10} x'_{10} + x_1^2 x'^2_1 + x_2^2 x'^2_2 + \cdots + x_{10}^2 x'^2_{10} + x_1^3 x'^3_1 + x_2^3 x'^3_2 + \cdots + x_{10}^3 x'^3_{10} \dots \\
 &= (\vec{x}^\top \vec{x}')^0 + (\vec{x}^\top \vec{x}')^1 + (\vec{x}^\top \vec{x}')^2 + (\vec{x}^\top \vec{x}')^3 + \dots \\
 &= \frac{1}{1 - \vec{x}^\top \vec{x}'}
 \end{aligned}$$

where the last line is because $\vec{x}^\top \vec{x}' \leq 1$ and therefore we can apply the formula for the infinite geometric series.

b. The kernel can be rewritten as

$$\begin{aligned}
 k(\vec{x}, \vec{x}') &= e^{-(\vec{x} - x_0)^\top (\vec{x} - x_0)} \\
 &= e^{-\vec{x}^\top \vec{x}} e^{2\vec{x}^\top \vec{x}'} e^{-\vec{x}'^\top \vec{x}'} \\
 &= e^{-\vec{x}^\top \vec{x}} \left(1 + \frac{2\vec{x}^\top \vec{x}'}{1!} + \frac{(2\vec{x}^\top \vec{x}')^2}{2!} + \frac{(2\vec{x}^\top \vec{x}')^3}{3!} + \dots \right) e^{-\vec{x}'^\top \vec{x}'} \\
 &= e^{-\vec{x}^\top \vec{x}} \left(1 + \frac{(\sqrt{2}\vec{x})^\top (\sqrt{2}\vec{x}')}{\sqrt{1!}\sqrt{1!}} + \frac{((\sqrt{2}\vec{x})^\top (\sqrt{2}\vec{x}'))^2}{\sqrt{2!}\sqrt{2!}} + \frac{((\sqrt{2}\vec{x})^\top (\sqrt{2}\vec{x}'))^3}{\sqrt{3!}\sqrt{3!}} + \dots \right) e^{-\vec{x}'^\top \vec{x}'}
 \end{aligned}$$

$$= e^{-\vec{x}^T \vec{x}} \begin{bmatrix} 1 \\ \sqrt{\frac{2}{1!}} x_1 \\ \sqrt{\frac{2}{1!}} x_2 \\ \vdots \\ \sqrt{\frac{2}{1!}} x_{10} \\ \sqrt{\frac{2^2}{2!}} x_1^2 \\ \sqrt{\frac{2^2}{2!}} x_2^2 \\ \vdots \\ \sqrt{\frac{2^2}{2!}} x_{10}^2 \\ \sqrt{\frac{2^3}{3!}} x_1^2 \\ \sqrt{\frac{2^3}{3!}} x_2^2 \\ \vdots \\ \sqrt{\frac{2^3}{3!}} x_{10}^2 \\ \vdots \end{bmatrix}^T \begin{bmatrix} 1 \\ \sqrt{\frac{2}{1!}} x'_1 \\ \sqrt{\frac{2}{1!}} x'_2 \\ \vdots \\ \sqrt{\frac{2}{1!}} x'_{10} \\ \sqrt{\frac{2^2}{2!}} x'^2_1 \\ \sqrt{\frac{2^2}{2!}} x'^2_2 \\ \vdots \\ \sqrt{\frac{2^2}{2!}} x'^2_{10} \\ \sqrt{\frac{2^3}{3!}} x'^2_1 \\ \sqrt{\frac{2^3}{3!}} x'^2_2 \\ \vdots \\ \sqrt{\frac{2^3}{3!}} x'^2_{10} \\ \vdots \end{bmatrix} e^{-\vec{x}'^T \vec{x}'} \quad \text{and thus } \phi(\vec{x}) = e^{-\vec{x}^T \vec{x}} \begin{bmatrix} 1 \\ \sqrt{\frac{2}{1!}} x_1 \\ \sqrt{\frac{2}{1!}} x_2 \\ \vdots \\ \sqrt{\frac{2}{1!}} x_{10} \\ \sqrt{\frac{2^2}{2!}} x_1^2 \\ \sqrt{\frac{2^2}{2!}} x_2^2 \\ \vdots \\ \sqrt{\frac{2^2}{2!}} x_{10}^2 \\ \sqrt{\frac{2^3}{3!}} x_1^2 \\ \sqrt{\frac{2^3}{3!}} x_2^2 \\ \vdots \\ \sqrt{\frac{2^3}{3!}} x_{10}^2 \\ \vdots \end{bmatrix}$$

Answer 7.8

- a. The update rule for logistic regression is

$$\vec{w} \leftarrow \vec{w} + \sum_{i=1}^m \eta (1 - \sigma(y_i \vec{x}_i^T \vec{w})) y_i \vec{x}_i$$

which is the sum of a set of terms, each of which is a positive scalar, $\eta(1 - \sigma(y_i \vec{x}_i^T \vec{w}))$, times y_i times a feature vector from the training set, $\vec{x}_i$. Thus, if we start with $\vec{w} = 0$ and apply this rule, each iteration will add some (positive) amount of each $y_i \vec{x}_i$ to $\vec{w}$ (and nothing else). So, the final result will also be some (non-negative) linear combination of $y_1 \vec{x}_1, \dots, y_m \vec{x}_m$.

- b. The total loss can be rewritten in terms of $a_1, \dots, a_m$ as

$$\begin{aligned} L &= \sum_{i=1}^m -\ln \sigma \left(y_i \vec{x}_i^T \sum_{h=1}^m a_h y_h \vec{x}_h \right) \\ &= \sum_{i=1}^m -\ln \sigma \left(\sum_{h=1}^m a_h y_i y_h \vec{x}_i^T \vec{x}_h \right) \\ &= \sum_{i=1}^m -\ln \sigma \left(\vec{a} \vec{G}_i \right) \end{aligned}$$

where the elements of G are $G_{ij} = y_i y_j \vec{x}_i^T \vec{x}_j$ and $\vec{G}_i$ is the i th row (or column) of G .

Therefore, the gradient is

$$\nabla_{\vec{a}} L = \sum_{i=1}^m - \left(1 - \sigma \left(\vec{a} \vec{G}_i \right) \right) \vec{G}_i$$

and the update rule is

$$\vec{a} \leftarrow \vec{a} + \eta \sum_{i=1}^m \left(1 - \sigma(\vec{a} \vec{G}_i)\right) \vec{G}_i$$

c. The data only appear in the above update rule in G . So, we just redefine the elements of G to be

$$G_{ij} = y_i y_j k(\vec{x}_i, \vec{x}_j) .$$

d. The discriminant in the linear case (replacing $\vec{w}$ with its expansion in terms of a_i) is

$$\begin{aligned} \tilde{f}(\vec{x}) &= \vec{x}^\top \sum_{h=1}^m a_h y_h \vec{x}_h \\ &= \sum_{h=1}^m a_h y_h \vec{x}^\top \vec{x}_h \end{aligned}$$

and so, replacing the inner product with the kernel, we get

$$\tilde{f}(xv) = \sum_{h=1}^m a_h y_h k(\vec{x}, \vec{x}_h)$$

or, to predict the probability of the positive class,

$$p(y = +1 \mid \vec{x}) = \sigma \left(\sum_{h=1}^m a_h y_h k(\vec{x}, \vec{x}_h) \right) .$$